

PLACEMENT PORTAL

A PROJECT REPORT

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**Under the supervision of
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CERTIFICATE

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ABSTRACT

The **Placement Portal** is a comprehensive digital platform designed to streamline the recruitment process for students, recruiters, and placement coordinators. The system aims to enhance accessibility, transparency, and efficiency in campus placements by providing a structured and automated environment for job applications, recruitment tracking, and document management.

For students, the portal offers an intuitive interface to create profiles, upload resumes, browse job postings, and apply for relevant opportunities. Recruiters can efficiently post job openings, filter applications based on criteria, and communicate with candidates. Placement coordinators play a pivotal role in managing the entire recruitment cycle, monitoring student applications, scheduling interviews, and ensuring smooth interaction between students and employers.

The platform integrates frontend technologies like HTML, CSS, and JavaScript, providing a user-friendly interface, while the backend is powered by **Django** with a database such as **MySQL** for efficient data management. Key features include **notifications for upcoming companies, application tracking, interview scheduling, Apply for jobs, Placement record management**.

By digitalizing the placement process, the system minimizes manual intervention, reduces paperwork, and ensures a seamless flow of information between all stakeholders. The expected outcome is an organized and automated placement management system that enhances student employability, facilitates recruiter engagement, and optimizes the efforts of placement coordinators. Ultimately, the Online Placement Portal fosters a transparent and efficient hiring ecosystem, improving the overall recruitment experience for educational institutions and companies alike.

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CHAPTER 1

INTRODUCTION

1.1 PROJECT DESCRIPTION

Placement Portal is a web-based application developed to streamline and automate the campus placement process for educational institutions. It provides a centralized system where students can register, view job opportunities, apply for positions, and participate in skill-enhancing events. Simultaneously, it allows administrators to manage student profiles, job postings, applications, and generate reports related to placements and company visits. By implementing role-based access control, it ensures secure and efficient management of placement activities for students, companies, and administrators. This project aims to digitize the traditional placement workflow, thereby enhancing transparency, accessibility, and speed in the recruitment process.

1.2 PROJECT SCOPE

- **Student Services**
 - Registration and profile management.
 - Viewing and applying for job postings.
 - Uploading resumes and tracking application status.
 - Receiving real-time notifications for job updates and events.
- **Administrative Functions**
 - Posting and managing job opportunities.
 - Accessing student applications and placement statuses.
 - Scheduling skill development events.
 - Generating placement reports and statistics.
- **Company Interaction**
 - Maintaining records of recruiting companies.
 - Viewing and updating results of ongoing recruitment drives.
 - Announcing upcoming company visits.
- **Security and Access Control**

- Role-based login for students, administrators, and company representatives.
- Ensures data privacy and restricted access based on user roles.
- **Reporting and Analysis**
 - Leaderboard showcasing highest and average packages.
 - Graphical statistics for company visits and historical placement data.
 - Summary of student placement distribution.

1.3 FUNCTIONAL REQUIREMENTS

1. User Registration and Authentication

- Students, admins, and companies can register/login using secure credentials.
- Role-based access (student, admin, company).

2. Student Module

- Register and manage their profile.
- Upload and update resumes.
- View job listings and apply for suitable opportunities.
- Track application status and receive notifications.
- Request For NOC.

3. Admin Module

- Add/edit/delete job postings.
- Approve or reject student applications.
- View and manage student records.
- Update and manage placement status.
- Schedule and publish upcoming events.
- Generate reports for placement statistics and student performance.
- Apporval for the NOC request.

4. Company Module

- Post job requirements.

- View student applications.
- Update recruitment results (selected/not selected).

5. Placement Reports

- View and generate reports:
 - Number of students placed/unplaced.
 - Company-wise selections.
 - Job role statistics.
 - Package analytics.

6. Event Scheduling

- Admin can create and manage skill-development events and sessions for students.

7. Search and Filter Options

- Job listings and company information can be searched and filtered based on role, package, location, etc.

1.4 NON- FONCTIONAL REQUIREMENTS

1. Performance

- The system should be able to handle multiple users simultaneously without significant delays.

2. Scalability

- Capable of accommodating a growing number of users and data as more students and companies participate.

3. Security

- Role-based access control.
- Secure password storage and authentication.

- Protection against unauthorized access and data leaks.

4. Usability

- User-friendly interface for both students and admins.
- Clear navigation and responsive design.

5. Availability

- The system should be accessible 24/7 with minimal downtime.

6. Reliability

- Ensures consistent performance and data integrity even after updates or in case of unexpected interruptions.

7. Compatibility

- Works across different devices and modern browsers (Chrome, Firefox, Edge, etc.).

8. Maintainability

- Code should be modular and well-documented to facilitate future updates and bug fixes.

9. Backup and Recovery

- Periodic backups of student data and job postings.
- Recovery mechanisms in case of system crash or data loss.

CHAPTER 2

FEASIBILITY STUDY

2.1 TECHNICAL FEASIBILITY

2.1.1. Concepts

- **Proof of Concept (PoC):**

Develop a basic version of the Placement Portal including user registration, job listing, and application tracking to validate core functionalities before proceeding to full-scale development.

Conduct testing with a small group of students and administrators to gather usability feedback and validate system workflow.

2.1.2 Infrastructure

- **Capacity and Performance:**

Evaluate the server environment (local/cloud) to ensure it can handle concurrent student and admin users during peak placement seasons.

Monitor response times and optimize for performance through caching and efficient database queries.

- **Functionality:**

Infrastructure should support role-based authentication, secure session management, real-time notifications, and file upload features (e.g., resume uploads).

2.1.3. Facilities

- **Data Center Requirements:**

If hosted on a physical server, ensure high availability, backup systems, and security controls are in place.

For cloud hosting, select providers offering reliable uptime and disaster recovery plans.

2.1.4. Architecture & Design

- **System Architecture:**

Adopt a layered architecture (frontend, backend, database) using Django for backend and HTML/CSS/JS for frontend to ensure scalability and maintainability.

Modular code structure to allow easy updates and future expansion (e.g., integrating AI-based placement suggestions).

- **Design Review:**

Perform peer and mentor reviews of both the user interface and backend logic.

Ensure the UI is responsive, accessible, and student-friendly.

2.1.5. Data

- **Data Quality Assessment:**

Ensure accurate and structured collection of student information, job postings, company details, and placement records.

Enforce input validation and secure handling of sensitive data (e.g., resumes, contact details).

2.1.6. Platform

- **Evaluation of Platforms:**

These platforms are open-source, highly functional, and align with project goals.

2.1.7. Components

- **Component Testing:**

Develop prototypes of key modules such as login system, job application form, and admin dashboard.

Conduct unit tests for individual components to ensure stability and performance under normal conditions.

2.1.8. Integration

- **System Integration:**

Define clear interaction flow between components: Student Module, Admin Module,

Company Data, Reports, and Event Scheduling.

Ensure smooth data flow using APIs and secure data handling practices.

- **Testing Integration:**

Perform integration testing to confirm that job postings, application submissions, and status updates flow correctly across modules and user roles.

2.2 OPERATIONAL FEASIBILITY

2.2.1. User Journey and Experience

- **Registration Process:**

Students register with academic details, upload resumes, and gain access to job listings.

Admins register through backend configuration with access to student records and control panels.

- **Application Process:**

Students can browse jobs, filter by company or role, and apply with one click.

Admins can track and update application status.

- **Feedback Mechanism:**

Implement an optional feedback form for students post-interview to gather insights on the recruitment experience.

2.2.2. Support Structure

- **User Support:**

Design a support system for students and admins including email support and an FAQ/help center within the portal.

- **Training and Resources:**

Provide user manuals or short video tutorials for students and staff on portal usage and functionalities.

2.2.3. Operational Processes

- **Placement Coordination:**

Admins post job opportunities and students apply via the portal.

Companies send their results to admins, which are then updated in the student dashboard.

- **Event Management:**

Schedule workshops or seminars under the “Upcoming Events” module with RSVP capability.

2.2.6. Marketing and Outreach

- **Awareness Campaigns:**

Promote the platform within the college through digital notices, WhatsApp groups, and classroom announcements.

- **User Engagement:**

Notify users of success stories, company feedback, and placement drives to maintain engagement.

2.2.7. Risk Management

- **Identifying Risks:**

Risks include system downtimes during placements, incomplete student records, and misuse of admin privileges.

- **Monitoring and Evaluation:**

Track user activity, application rates, and placement success using internal analytics dashboards.

Regularly assess admin feedback for system improvements.

2.3 BEHAVIORAL FEASIBILITY

2.3.1. Stakeholders' Motivation

- **Students:**

Motivated by a centralized, transparent, and accessible system for applying to jobs and checking placement progress.

- **Administrators:**

Benefit from automated records, easy tracking, and reduced manual paperwork.

- **Companies:**

Gain a simplified process to access verified student data and recruitment outcomes.

2.3.2. Ease of Use

- **User Interface Design:**

Clean layout, intuitive navigation, and mobile-responsive design for accessibility.

- **Support and Contact:**

Clear help/contact options to guide users in case of queries or technical issues.

- **Behavioral Acceptance:**

High, due to relevance to academic needs and alignment with existing placement practices.

CHAPTER 3

PROJECT OBJECTIVES

3.1. Facilitate Student Engagement

To connect students with job opportunities effectively through an intuitive and centralized portal.

- Allow students to create and manage detailed profiles, including academic performance, skills, and uploaded resumes.
- Enable students to browse, filter, and apply to job postings from multiple companies.
- Simplify communication between students and placement coordinators through in-portal notifications and updates.
- Include a personalized dashboard for students to track applications, interview schedules, and placement status.

3.2. Empower Placement Administrators

To provide tools and automation that streamline placement office operations.

- Provide an admin dashboard to post jobs, manage student applications, and update placement results.
- Offer analytics features to generate reports on placement statistics, student participation, and company recruitment trends.
- Implement event management features for scheduling skill-development sessions and workshops.
- Maintain a historical database of placements for future reference and analysis.

3.3. Bridge the Gap with Recruiting Companies

To make company-student interactions seamless and efficient.

- Store and display information about recruiting companies, including eligibility criteria and selection processes.
- Enable companies to view shortlisted students and upload results directly or via admin support.

- Provide real-time updates to students on the status of company visits and interview outcomes.
- Maintain a record of past and upcoming company visits with related outcomes.

3.4. Promote Transparency and Accessibility

To ensure equal access and clear visibility of placement activities across the institution.

- Use role-based login systems to secure user data while offering appropriate access levels to students, admins, and companies.
- Display real-time placement statistics like number of students placed, average packages, and top recruiters.
- Provide clear timelines and deadlines for application submissions and event participation.
- Maintain an accessible interface for mobile and desktop users to promote maximum participation.

CHAPTER 4

PROJECT OUTCOMES

Development of a Centralized Placement Platform

A fully functional and user-friendly web-based platform was successfully developed to serve as a unified solution for managing campus placements. The portal enables students to apply for jobs, administrators to oversee placement activities, and companies to share opportunities, all from a single platform.

• Transparent Placement Process and Real-time Updates

The system introduced transparency in the placement workflow by enabling real-time tracking of applications and placement statuses. Students receive live notifications about job openings, selection updates, and interview schedules, ensuring they are always informed.

• Role-based User Access and Secure Authentication

The platform supports custom registration and secure login for different user roles: students, administrators, and companies. Each role is assigned specific permissions — students can apply for jobs, admins can manage records and post jobs, while companies can access relevant student data and recruitment results.

• Comprehensive Job and Student Management

Administrators can efficiently manage job postings, company details, and student applications. Students can easily view job listings, apply with uploaded resumes, and track their progress through the placement cycle. The system simplifies the overall placement process for both students and admins.

• Responsive and Intuitive User Interface

The portal was built with a mobile-responsive and user-centric design, ensuring accessibility across desktops, tablets, and mobile devices. Clean navigation, organized modules, and consistent layout contribute to a positive and productive user experience.

- **Modular Design Supporting Future Scalability**

The project followed a modular architecture with distinct components for student management, job listings, company interactions, and event scheduling. This design facilitates easy maintenance and supports future features like resume analyzers, AI-based matching, and mobile app integration.

- **Future-readiness and Scope for Expansion**

The platform is ready for future upgrades such as integration with third-party recruitment tools, analytics dashboards for placement trends, and a dedicated mobile app. Its scalable backend ensures it can handle a growing user base and evolving institutional needs.

- **Enhanced Placement Support and Institutional Efficiency**

The Placement Portal has significantly improved placement-related operations by reducing manual workload, increasing data accessibility, and enabling timely actions. It strengthens institutional capabilities and improves students' placement experiences by offering a centralized and well-organized ecosystem.

4.1 USER INTERFACE DESIGN

- **4.1.1. Landing Page**

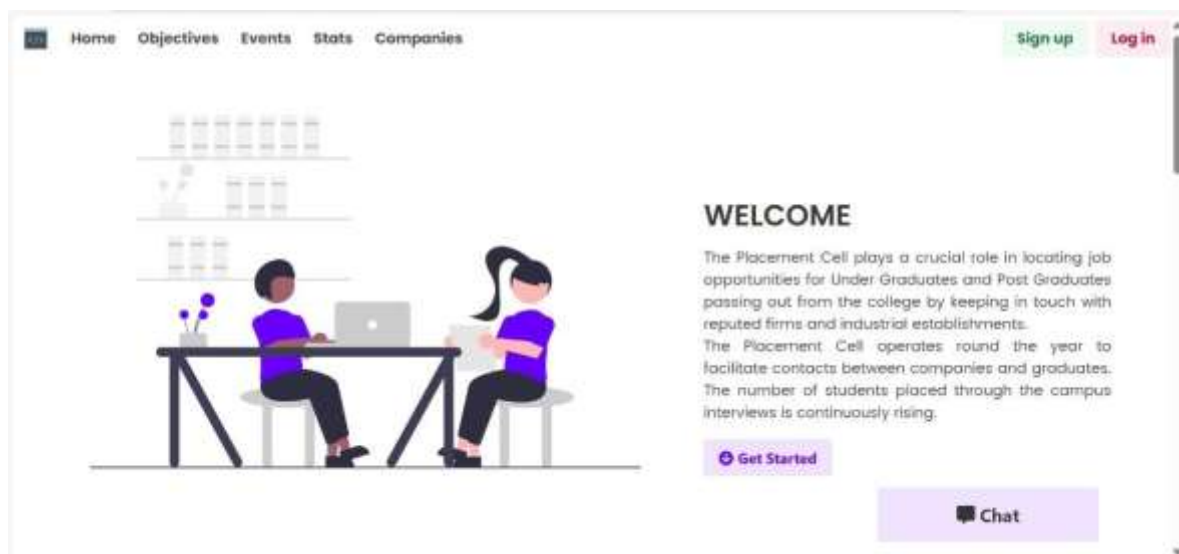
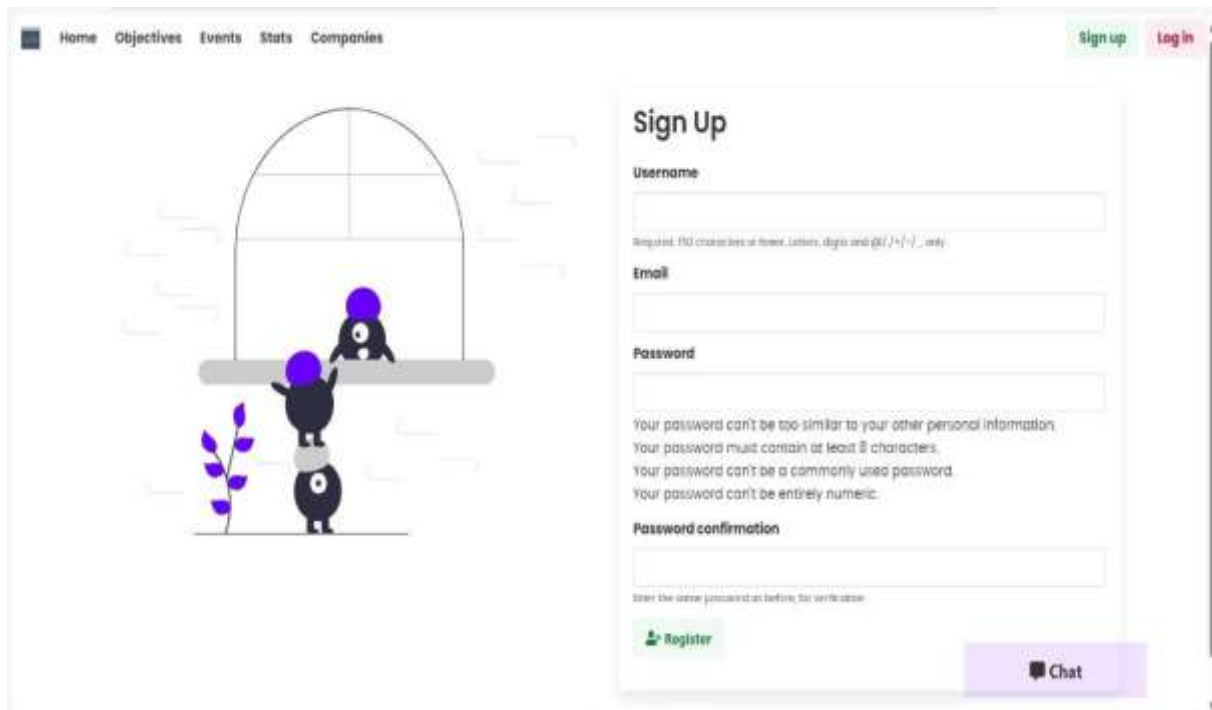


Fig. 4.1: Landing page

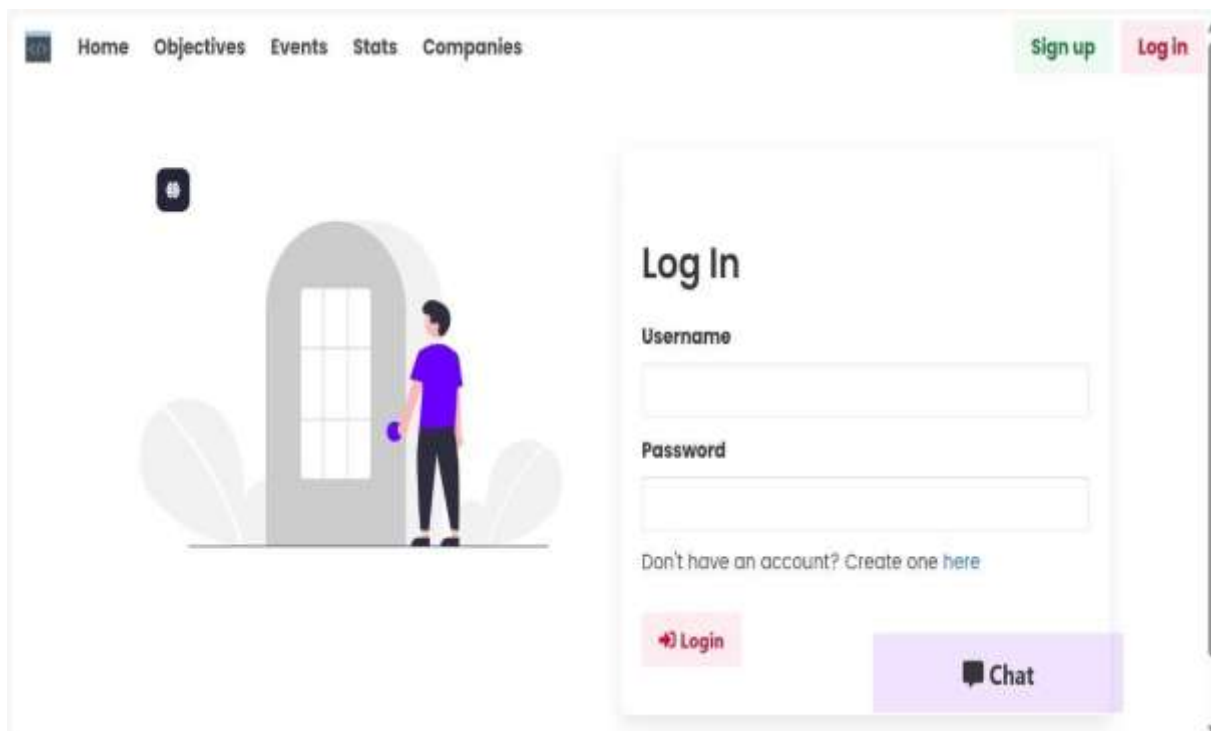
- **4.1.2. Signup Page**



The screenshot shows a web application interface for a signup page. At the top, there is a navigation bar with links: Home, Objectives, Events, Stats, and Companies. In the top right corner, there are buttons for 'Sign up' (green) and 'Log in' (pink). The main content area is split into two sections. On the left, there is an illustration of two small, round, purple characters with large eyes; one is standing and the other is sitting on a shelf under a large, arched window. On the right, there is a 'Sign Up' form. The form has four input fields: 'Username', 'Email', 'Password', and 'Password confirmation'. Below the 'Username' field, there is a small text note: 'Required: 150 characters of lower, letters, digits and @!\$%^&*~`-_, only'. Below the 'Password' field, there are four lines of text: 'Your password can't be too similar to your other personal information.', 'Your password must contain at least 8 characters.', 'Your password can't be a commonly used password.', and 'Your password can't be entirely numeric.' Below the 'Password confirmation' field, there is a small text note: 'Enter the same password as before, for verification'. At the bottom of the form, there is a green 'Register' button with a small icon of a person. To the right of the form, there is a purple 'Chat' button with a speech bubble icon.

Fig. 4.2: Signup page

- **4.1.3. Login Page**



The screenshot shows a web application interface for a login page. At the top, there is a navigation bar with links: Home, Objectives, Events, Stats, and Companies. In the top right corner, there are buttons for 'Sign up' (green) and 'Log in' (pink). The main content area is split into two sections. On the left, there is an illustration of a person with dark hair, wearing a purple shirt and black pants, standing next to a large, arched doorway. On the right, there is a 'Log In' form. The form has two input fields: 'Username' and 'Password'. Below the 'Password' field, there is a link: 'Don't have an account? Create one here'. At the bottom of the form, there is a pink 'Login' button with a small icon of a person. To the right of the form, there is a purple 'Chat' button with a speech bubble icon.

Fig. 4.3: Login page

- 4.1.4. Companies

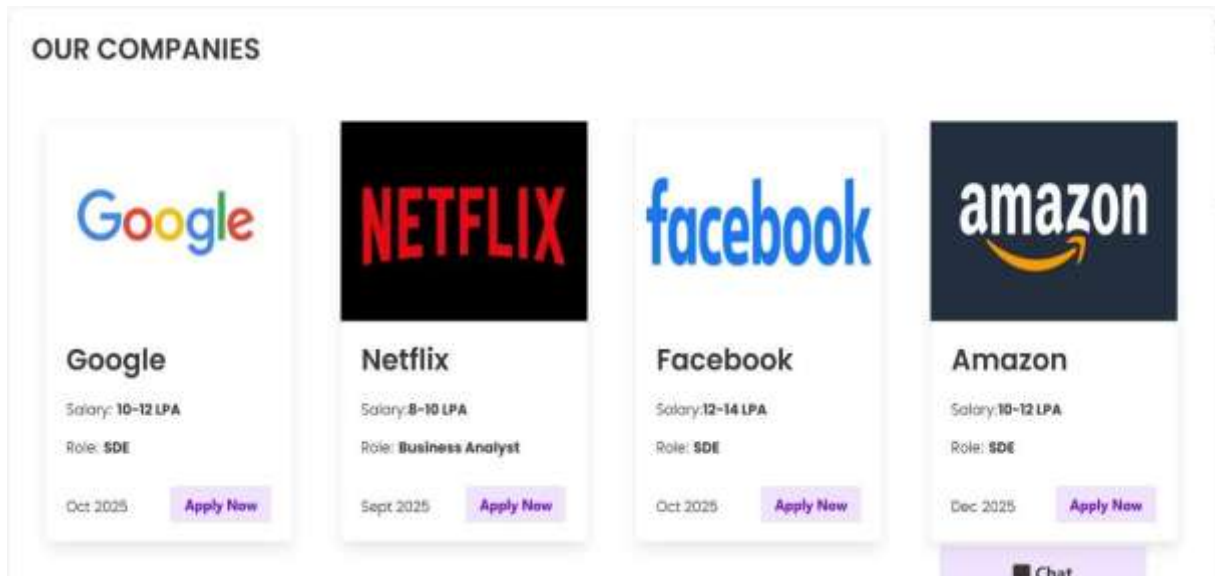


Fig. 4.4: Companies page

- 4.1.5. Upcoming Companies

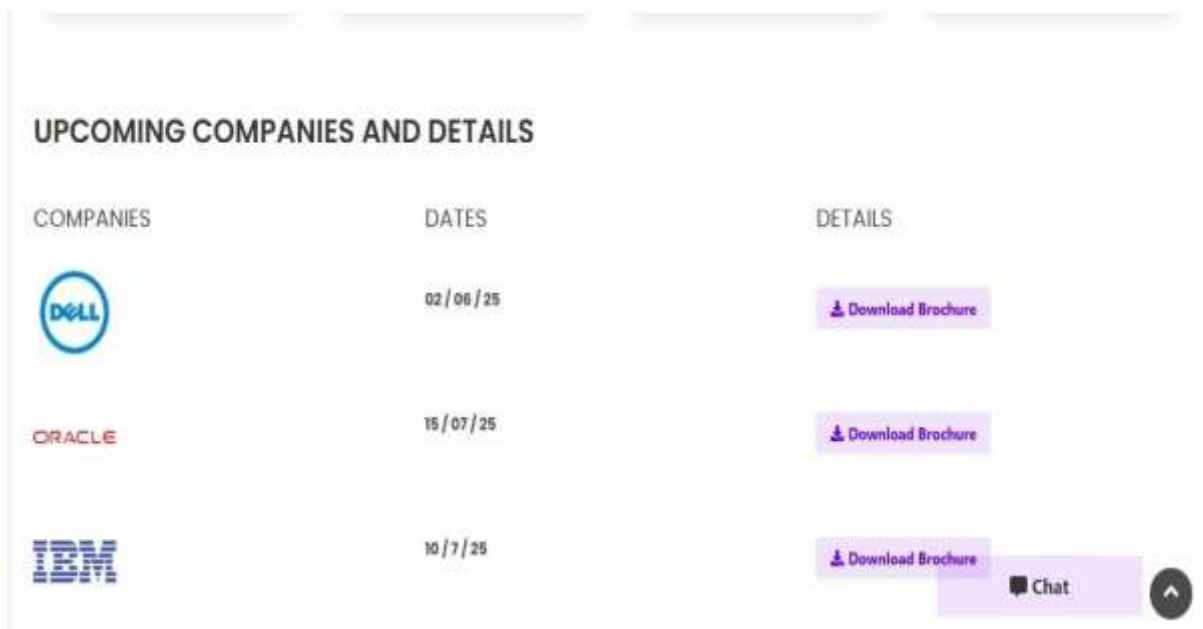


Fig. 4.5: Upcoming Companies page

- **4.1.6. Applications**

Home Objectives Events Stats Companies

admin logout

Apply for job:

Username:

Email id:

Phone no:

College:

Graduation Year:

Company:

Job profile:

Fig. 4.6: Apply for job page

- **4.1.7. Statistics**



Fig. 4.7: Statistics page

- **4.1.9. Events Register**

Fig. 4.8: Event Register page

Name	Purpose	Status	Date & Time	Proof
sath	Placement	Approved	8/7/2025, 11:23:10 am	View
divya	Internship	Approved	7/5/2025, 10:58:16 am	View

Fig. 4.9: NOC Request Form page

4.2 MODULE DESCRIPTION

Module 1: Student Module

This module facilitates student interaction with the placement portal and helps them manage their placement journey. It includes the following functionalities:

- **Account Setup:**
Students can register and create profiles by entering academic details, contact information, skills, and uploading their resumes.
- **Job Application:**
Students can view job listings posted by companies or administrators, filter them based on eligibility or interest, and apply directly through the portal.
- **Application Tracking:**
Students can track the status of their applications — whether shortlisted, selected, or rejected — in real time via their dashboard.
- **Event Participation:**
Students can register for skill-enhancement sessions, webinars, and placement-related events organized by the college.

Module 2: Admin Module

This module empowers placement administrators to manage all backend operations and ensures a smooth placement process. It includes:

- **Job Posting Management:**
Admins can add, edit, or delete job postings and company information. They can assign eligibility criteria, deadlines, and interview schedules.
- **Student Application Tracking:**
Admins can access applications submitted by students, track recruitment progress, and update placement statuses.
- **Placement Reports:**
Admins can generate detailed reports on placed/unplaced students, company participation, and package statistics.
- **Event and Notification Management:**
Admins can create events for students and send out notifications or alerts about job openings, deadlines, and results.

Module 3: Company Module

This module allows company representatives to participate in the recruitment process efficiently. It includes:

- **Company Profiles:**
Companies can register and manage their profiles, including adding job roles, eligibility details, and interview stages.
- **Application Review:**
Companies can review applications, shortlist candidates, and share results with the admin through the portal.
- **Recruitment Status Updates:**
Companies can update the outcome of their recruitment drive, which is reflected in the student and admin dashboards.

Module 4: Statistics and Reports

This module is designed to offer data insights and analytics related to placement performance.

- **Placement Dashboard:**
Displays real-time stats including the number of students placed, average and highest packages, and top recruiters.
- **Leaderboards:**
Graphs and tables display company-wise placement figures, salary trends, and placement history for strategic analysis.

Module 5: Upcoming Events & Workshops

This module enhances student preparedness by enabling participation in various development events.

- **Event Scheduling:**
Admins can publish and manage technical sessions, soft skill workshops, and interview training events.

- **Student Registration:**

Students can view event details, register for sessions, and receive reminders and updates via the portal.

Module 6: Security & Role-Based Access

This module ensures secure access to the system and protects sensitive data.

- **Authentication:**

Secure login and session management for students, admins, and company users.

- **Access Control:**

Each role has predefined permissions, ensuring that users only see or interact with features relevant to their role.

4.3 DATABASE DESIGN

4.3.1. Use Case Diagram

The use case diagram outlines the interactions between different users (students, admins, and companies) and the Placement Portal system, showcasing the primary functionalities offered.

Actors:

- **Student:** A user (candidate) looking for placement opportunities.
- **Admin:** The administrator managing users, jobs, and placement statistics.
- **Company:** The recruiter posting jobs and accessing student applications.

Use Cases:

Student:

- Register and Login
- Fill Placement Form
- View Job Openings
- Apply for Jobs
- View Application Status
- Track Placement Status

Company:

- Register/Login
- Post Job Requirements
- View Student Applications
- Shortlist Candidates
- Schedule Interviews

Admin:

- Manage Student and Company Accounts
- Approve/Reject Registrations
- Add/Edit/Delete Job Posts
- View Placement Statistics (Placed/Unplaced Count)
- Generate Reports
- Handle Notifications

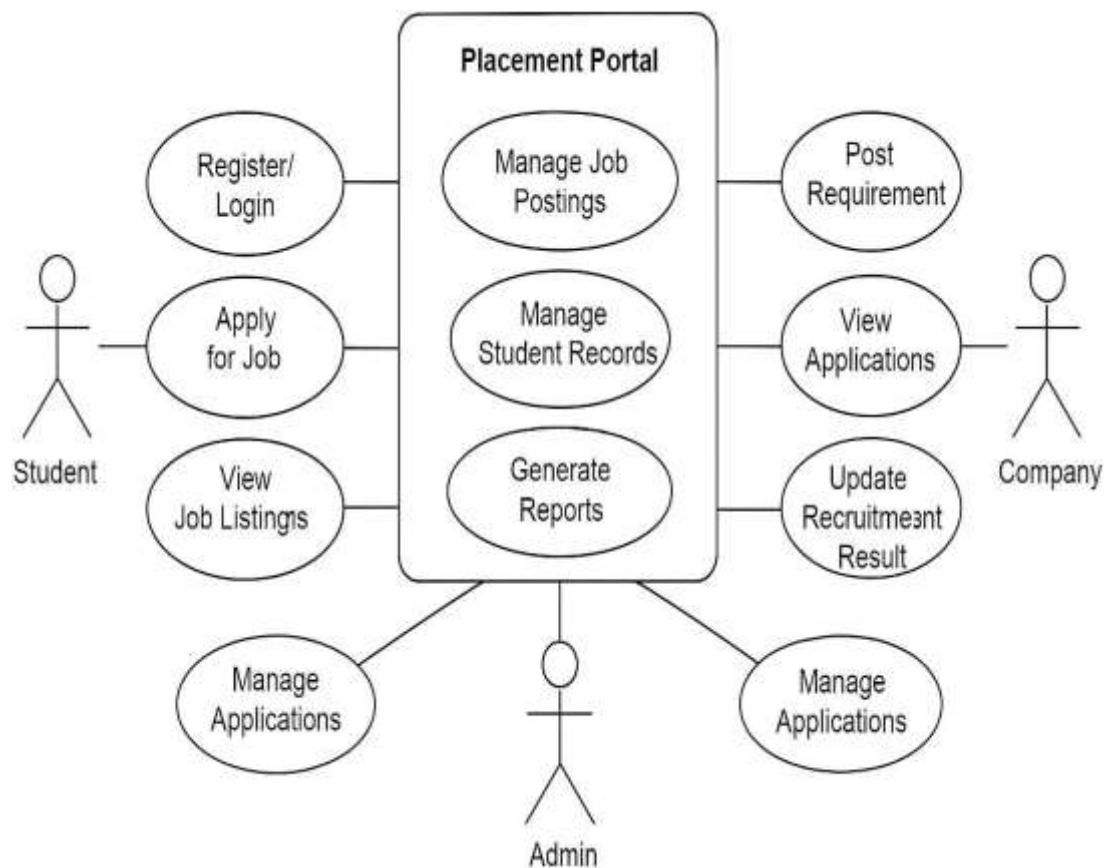


Fig 4.9: Use case diagram (Placement Portal)

4.3.2. Entity Relationship Diagram

- ER model stands for an Entity-Relationship model. It is a high-level data model. This model is used to define the data elements and relationship for a specified system.
- It develops a conceptual design for the database. It also develops a very simple and easy to design view of data.
- In ER modelling, the database structure is portrayed as a diagram called an entityrelationship diagram.

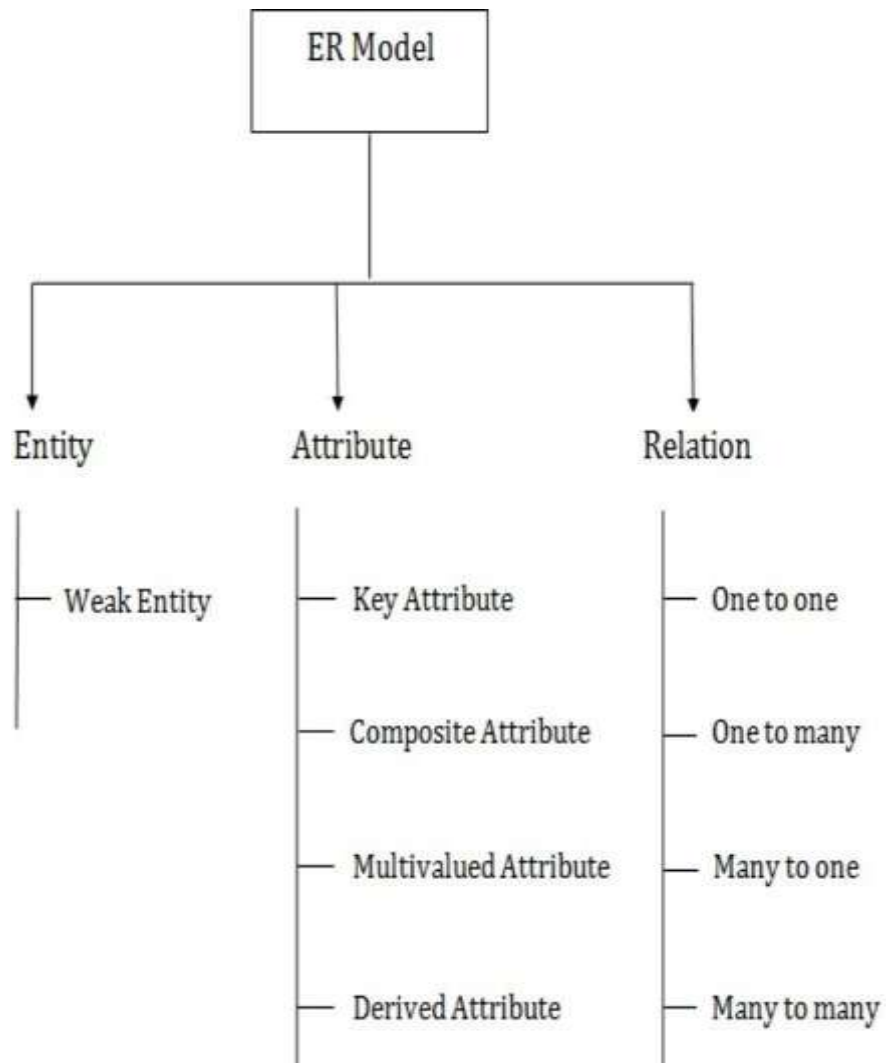


Fig. 4.10: E-R model

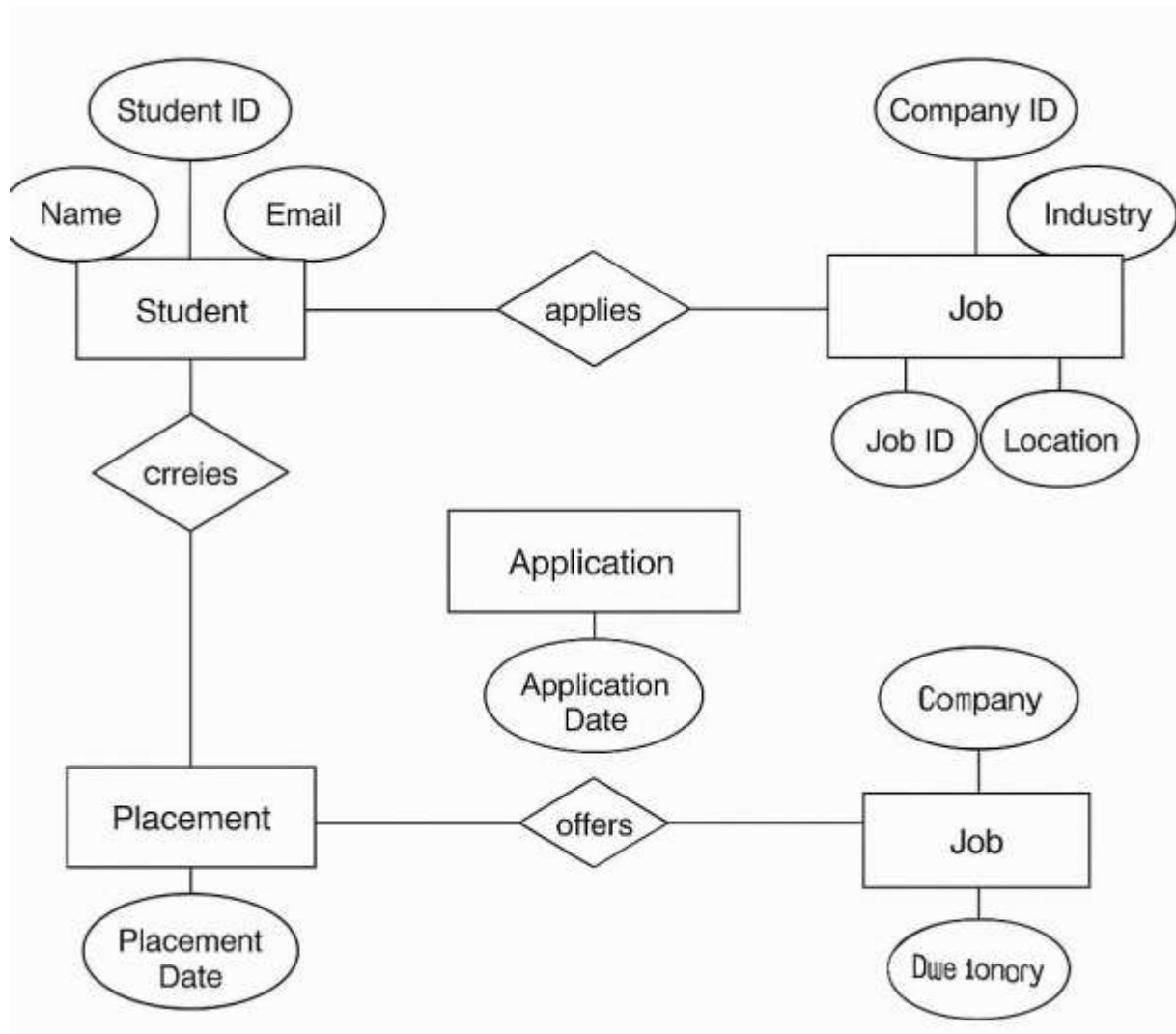


Fig. 4.11: E-R diagram(Placement Portal)

4.4 SUMMARY

The Placement Portal is a web-based system developed to improve and automate the campus placement process for educational institutions. It provides a centralized platform where students can register, create profiles, upload resumes, view job listings, apply for positions, and participate in skill-enhancement events. Administrators can manage job postings, monitor student applications, update placement statuses, and generate reports, all through a secure, role-based access system that ensures data privacy and integrity.

The system draws inspiration from existing Training & Placement Cell portals and third-party platforms such as LinkedIn, Naukri, and Internshala. It incorporates essential features like student profile management, job application tracking, and notifications for deadlines and updates. Developed using HTML, CSS, JavaScript for the frontend and Django for the backend, the portal uses MySQL or SQLite as its database, ensuring reliable performance and ease of integration.

Key modules include the Student Module for job applications, the Admin Module for managing records, a Statistics Module with leaderboards and graphs, an Events Module for scheduled activities, and a Companies Module for information on recruiters. These modules work together to create a seamless and efficient placement experience.

The system also provides visual tools like Gantt charts and workflows to map the development and operational process. Reports generated from the portal offer insights into student placements, job postings, and company visits.

In summary, the Placement Portal is an effective, secure, and scalable solution that benefits both students and administrators by simplifying placement procedures and enhancing the overall recruitment process.

CHAPTER 5

HARDWARE AND SOFTWARE REQUIREMENTS

5.1. HARDWARE REQUIREMENTS

These are the essential physical components required to develop, deploy, and run the Placement Portal system efficiently.

- **Server Requirements:**

- **Processor:** Multi-core processor (Intel i5/i7 or AMD Ryzen equivalent)
- **RAM:** Minimum 8 GB (Recommended: 16 GB for handling multiple user sessions)
- **Storage:** Solid State Drive (SSD) with at least 100 GB capacity to store logs, system files, resumes, and reports
- **Network:** High-speed internet connection with sufficient bandwidth to support simultaneous access by students, admins, and companies
- **Backup Device:** External hard drive or cloud-based storage for automated periodic backups of system data and user records

- **User-side Requirements:**

- **Device:** Compatible desktop, laptop, tablet, or smartphone
- **Browser:** Modern web browser such as Google Chrome, Mozilla Firefox, Safari, or Microsoft Edge with support for HTML5, CSS3, and JavaScript
- **Internet:** Reliable internet connection to ensure smooth navigation and data transmission

5.2. SOFTWARE REQUIREMENTS

These include the development frameworks, tools, and platforms necessary for building and maintaining the Placement Portal.

- **Development Tools:**

- **Frontend:** HTML, CSS, JavaScript (for structure and interactivity of the web interface)
- **Backend:** Django (Python-based framework for server-side development)

- **Database:** MySQL or SQLite (If Applicable in future).
- **Version Control:** Git and GitHub (for source code versioning and collaboration)
- **IDE/Code Editor:** Visual Studio Code (for development and debugging)
- **Browser Compatibility Testing:** Chrome DevTools, Firefox Developer Edition

CHAPTER 6

REFERENCES

1. **Sommerville, I. (2011).** *Software Engineering* (9th Edition). Pearson Education. — Provided foundational knowledge on software engineering principles, modular system architecture, and software development methodologies applicable to building the Placement Portal system.
2. **Pressman, R. S. (2014).** *Software Engineering: A Practitioner's Approach* (8th Edition). McGraw-Hill Education. — Used for understanding requirement analysis, system modeling, and design processes relevant to managing placement workflows.
3. **Django Official Documentation.** — Referred extensively for backend development, including user authentication, ORM (Object Relational Mapping), and integration of admin functionalities.
<https://docs.djangoproject.com>
4. **Mozilla Developer Network (MDN) Web Docs.** — Used to reference standard practices in HTML5, CSS3, and JavaScript for building responsive and user-friendly front-end components.
<https://developer.mozilla.org/en-US/>
5. **W3Schools.** *Web Development Tutorials.* — Utilized for learning and implementing web design fundamentals, form handling, and UI/UX best practices.
<https://www.w3schools.com/tutorials/>
6. **MySQL Documentation.** — Consulted for database structure, query optimization, and connection handling in managing job postings, student applications, and placement records.
<https://dev.mysql.com/doc/>
7. **GitHub Documentation.** — Used for version control and collaborative development across the project team using Git and GitHub.
<https://docs.github.com/>